

Deep Learning Project Report

This project implements a small deep learning image classification model using PyTorch and transfer learning. The model is trained on the CIFAR-10 dataset and uses a pretrained ResNet architecture for improved accuracy.

Project Goal

Implement a deep learning image classification model using transfer learning and evaluate its performance on the CIFAR-10 dataset.

Dataset

Dataset Used: CIFAR-10

Number of Classes: 10

Classes include airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck.

Technologies Used

- Python
- PyTorch
- TorchVision
- Matplotlib
- Scikit-learn

Model Architecture

The project uses a pretrained ResNet18 model from TorchVision. The final fully connected layer is modified to classify CIFAR-10 images into 10 classes.

Data Augmentation

The following augmentation techniques are applied:

- Random Horizontal Flip
- Random Rotation
- Normalization

Training Process

1. Load and preprocess CIFAR-10 dataset.
2. Apply data augmentation.
3. Split data into training and validation sets.

4. Train pretrained ResNet18 model.
5. Evaluate performance using classification metrics.

Evaluation Metrics

The following metrics are used:

- Accuracy
- Precision
- Recall
- F1-score

Training Curves

Training and validation loss curves are plotted using Matplotlib to visualize model learning performance.

Model Saving

The trained model is saved as:

model.pth

Inference Instructions

Run the following commands:

```
python train.py
python predict.py
```

Useful Links

PyTorch Transfer Learning Tutorial:
https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html

CIFAR-10 Dataset:
<https://www.cs.toronto.edu/~kriz/cifar.html>

TorchVision Models:
<https://pytorch.org/vision/stable/models.html>

Conclusion

This project demonstrates how transfer learning can improve image classification performance even on small datasets. Pretrained models help reduce training time and improve accuracy.